

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460356.83341 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1929 +/- 0.008
Transit depth (Rp/Rs)^2	3.72 +/- 0.31 [%]
Orbital Inclination (inc)	80.83 +/- 0.49
Airmass coefficient 1 (a1)	1.254 +/- 0.0093
Airmass coefficient 2 (a2)	-0.162 +/- 0.03
Scatter in the residuals of the lightcurve fit is	0.74 %
Best Comparison Star	None
Optimal Aperture	4.23
Optimal Annulus	10.39
Transit Duration (day)	0.0491 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1929 $\pm$ 0.008 vs 0.1594 (deviation 3.82 σ)
Duration fit vs published	0.0491 d vs 0.0512 d (deviation 0.98 σ)
Mid-time O-C	-5.6 min
Depth SNR	22.0
Residual scatter	0.74 %

Light curve

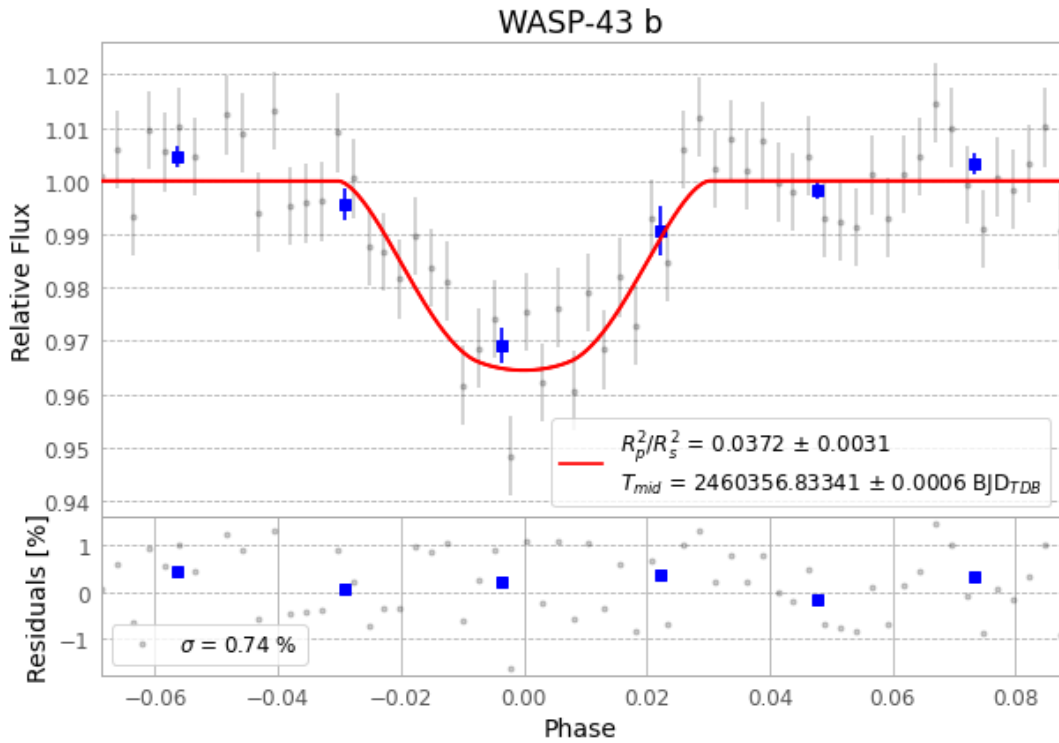


Figure 1 — FinalLightCurve_WASP-43 b_2024-02-15.png (SHA-256 0edb8a3957b235cc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [488, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 836c4526b089f4d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

62 FITS frames (41 MB), WASP-43240216063015.FITS → WASP-43240216093315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240216.log (309a0cab9d70...), FinalLightCurve_WASP-43 b_2024-02-15.png (0edb8a3957b2...), FinalLightCurve_WASP-43 b_2024-02-15.csv (247e97a8e0b0...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 18:17Z.